
1. BASIC ROUTER / SWITCH INITIAL CONFIGURATION & INTERFACE SETUP

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet 0/0
Router(config-if)# ip address 192.168.10.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# do write memory
```

2. LINE PASSWORDS & ENABLE SECRET SECURITY

```
Router> enable
Router# configure terminal

! Configure Console Line Password
Router(config)# line console 0
Router(config-line)# password abc
Router(config-line)# login
Router(config-line)# exit
```

```
! Configure Enable Secret Password
Router(config)# enable secret abc
Router(config)# exit
```

```
! Verify Running Configuration
Router# show running-config
```

3. REMOTE ACCESS (TELNET CONFIGURATION)

```
Router> enable
Router# configure terminal
Router(config)# line vty 0 4
Router(config-line)# password abc
Router(config-line)# login
Router(config-line)# exit
Router(config)# do write memory
```

4. DHCP SERVER CONFIGURATION

```
Router> enable
Router# configure terminal
Router(config)# ip dhcp pool My_LAN_Pool
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
Router(config)# do write memory
```

5. STATIC ROUTING

```
Router> enable
Router# configure terminal
! Syntax: ip route <destination-network> <subnet-mask> <next-hop-ip-or-exit-int>
Router(config)# ip route 192.168.30.0 255.255.255.0 192.168.70.254
Router(config)# do write memory
```

6. DYNAMIC ROUTING: RIP VERSION 2 (IPv4)

```
Router1> enable
Router1# configure terminal
Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# network 192.168.1.0
Router1(config-router)# network 192.168.10.0
Router1(config-router)# network 192.168.20.0
Router1(config-router)# exit
Router1(config)# do write memory
Router1(config)# do show ip route
```

7. DYNAMIC ROUTING: EIGRP (IPv4)

```
Router1> enable
Router1# configure terminal
Router1(config)# router eigrp 10
Router1(config-router)# network 192.168.1.0
Router1(config-router)# network 192.168.10.0
Router1(config-router)# network 192.168.20.0
Router1(config-router)# exit
```

```
Router1(config)# do write memory
Router1(config)# do show ip route
```

8. DYNAMIC ROUTING: OSPF (IPv4 3-ROUTER TOPOLOGY)

! Router 1 Configuration

```
Router1> enable
Router1# configure terminal
Router1(config)# router ospf 1
Router1(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router1(config-router)# network 10.1.1.0 0.0.0.3 area 0
Router1(config-router)# exit
Router1(config)# do write memory
```

! Router 2 Configuration

```
Router2> enable
Router2# configure terminal
Router2(config)# router ospf 1
Router2(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router2(config-router)# network 10.1.1.0 0.0.0.3 area 0
Router2(config-router)# network 10.2.2.0 0.0.0.3 area 0
Router2(config-router)# exit
Router2(config)# do write memory
```

! Router 3 Configuration

```
Router3> enable
Router3# configure terminal
Router3(config)# router ospf 1
Router3(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router3(config-router)# network 10.2.2.0 0.0.0.3 area 0
Router3(config-router)# exit
Router3(config)# do write memory
```

9. ACCESS CONTROL LISTS (ACL) - GENERAL SYNTAX & CONFIGURATION TASKS

! =====

! ACL GENERAL SYNTAX REFERENCE

! =====

! 1. Standard Numbered ACL Syntax (ID Range: 1 - 99):

! Router(config)# access-list <1-99> {permit|deny} <source-ip> [wildcard-mask]

! Router(config)# access-list <1-99> {permit|deny} host <source-ip>

```
! Router(config)# access-list <1-99> {permit|deny} any
!
! 2. Extended Numbered ACL Syntax (ID Range: 100 - 199):
! Router(config)# access-list <100-199> {permit|deny} <protocol> <source-ip>
<source-wildcard> <dest-ip> <dest-wildcard> [eq port]
! Note: <protocol> can be ip, tcp, udp, icmp, etc.
! Note: Common port numbers / keywords: 23 (telnet), 80 (www/http), 21 (ftp), 53
(domain/dns)
!
! 3. Named ACL Syntax (Standard or Extended):
! Router(config)# ip access-list {standard | extended} <acl-name>
! Router(config-std-nacl)# {permit | deny} <source-ip> [wildcard-mask]
! Router(config-ext-nacl)# {permit | deny} <protocol> <source> <source-wildcard>
<destination> <dest-wildcard> [eq port]
!
! 4. Applying ACL to an Interface:
! Router(config-if)# ip access-group <acl-number-or-name> {in | out}
!
! 5. Verification & Management Commands:
! Router# show access-lists
! Router# show ip interface <interface-id>
```

```
! =====
! PRACTICAL LAB CONFIGURATION TASKS
! =====
```

! Task 1: Standard ACL - Block a particular host from accessing all connected
! devices on the second router.

! Note: Standard ACLs filter based on source IP and are placed close to destination.

```
Router2> enable
Router2# configure terminal
Router2(config)# access-list 10 deny host 192.168.10.11
Router2(config)# access-list 10 permit any
Router2(config)# interface FastEthernet 0/0
Router2(config-if)# ip access-group 10 out
Router2(config-if)# exit
Router2(config)# do write memory
```

! Task 2: Extended ACL - Block one entire network from accessing another network.

! Note: Extended ACLs filter source to destination and are placed close to source.

```
Router1> enable
Router1# configure terminal
Router1(config)# access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
```

```
Router1(config)# access-list 100 permit ip any any
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ip access-group 100 in
Router1(config-if)# exit
Router1(config)# do write memory
```

! Task 3: Extended ACL - Block Telnet (TCP 23) and HTTP (TCP 80) between two networks.

```
Router1> enable
Router1# configure terminal
Router1(config)# access-list 101 deny tcp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 23
Router1(config)# access-list 101 deny tcp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 80
Router1(config)# access-list 101 permit ip any any
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ip access-group 101 in
Router1(config-if)# exit
Router1(config)# do write memory
```

10. NETWORK ADDRESS TRANSLATION (NAT) - 2 ROUTERS CONFIGURATION

! Task 1: Static NAT on Two Routers (Fixed 1-to-1 Mapping)

! Router 1 Setup

```
Router1> enable
Router1# configure terminal
Router1(config)# ip nat inside source static 192.168.10.11 200.1.1.10
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ip nat inside
Router1(config-if)# exit
Router1(config)# interface Serial 0/0/0
Router1(config-if)# ip nat outside
Router1(config-if)# exit
Router1(config)# do write memory
```

! Router 2 Setup

```
Router2> enable
Router2# configure terminal
Router2(config)# ip nat inside source static 192.168.20.11 200.1.1.20
Router2(config)# interface FastEthernet 0/0
Router2(config-if)# ip nat inside
Router2(config-if)# exit
Router2(config)# interface Serial 0/0/0
Router2(config-if)# ip nat outside
Router2(config-if)# exit
```

Router2(config)# do write memory

! Task 2: Dynamic NAT on Two Routers (Pool Translation)

! Router 1 Setup

Router1> enable

Router1# configure terminal

Router1(config)# access-list 10 permit 192.168.10.0 0.0.0.255

Router1(config)# ip nat pool NAT_POOL1 200.1.1.100 200.1.1.110 netmask 255.255.255.0

Router1(config)# ip nat inside source list 10 pool NAT_POOL1

Router1(config)# interface FastEthernet 0/0

Router1(config-if)# ip nat inside

Router1(config-if)# exit

Router1(config)# interface Serial 0/0/0

Router1(config-if)# ip nat outside

Router1(config-if)# exit

Router1(config)# do write memory

! Router 2 Setup

Router2> enable

Router2# configure terminal

Router2(config)# access-list 10 permit 192.168.20.0 0.0.0.255

Router2(config)# ip nat pool NAT_POOL2 200.1.1.200 200.1.1.210 netmask 255.255.255.0

Router2(config)# ip nat inside source list 10 pool NAT_POOL2

Router2(config)# interface FastEthernet 0/0

Router2(config-if)# ip nat inside

Router2(config-if)# exit

Router2(config)# interface Serial 0/0/0

Router2(config-if)# ip nat outside

Router2(config-if)# exit

Router2(config)# do write memory

! Task 3: PAT (Port Address Translation / Overload) on Two Routers

! Router 1 Setup

Router1> enable

Router1# configure terminal

Router1(config)# access-list 10 permit 192.168.10.0 0.0.0.255

Router1(config)# ip nat inside source list 10 interface Serial 0/0/0 overload

Router1(config)# interface FastEthernet 0/0

Router1(config-if)# ip nat inside

Router1(config-if)# exit

Router1(config)# interface Serial 0/0/0

Router1(config-if)# ip nat outside

Router1(config-if)# exit

```
Router1(config)# do write memory
```

```
! Router 2 Setup
```

```
Router2> enable
```

```
Router2# configure terminal
```

```
Router2(config)# access-list 10 permit 192.168.20.0 0.0.0.255
```

```
Router2(config)# ip nat inside source list 10 interface Serial 0/0/0 overload
```

```
Router2(config)# interface FastEthernet 0/0
```

```
Router2(config-if)# ip nat inside
```

```
Router2(config-if)# exit
```

```
Router2(config)# interface Serial 0/0/0
```

```
Router2(config-if)# ip nat outside
```

```
Router2(config-if)# exit
```

```
Router2(config)# do write memory
```

11. IPV6 ROUTING CONFIGURATION

```
! Task 1: Configure RIP for IPv6 (RIPng) on Two Routers
```

```
! Router 1 Configuration
```

```
Router1> enable
```

```
Router1# configure terminal
```

```
Router1(config)# ipv6 unicast-routing
```

```
Router1(config)# ipv6 router rip MY_RIP
```

```
Router1(config-rtr)# exit
```

```
Router1(config)# interface FastEthernet 0/0
```

```
Router1(config-if)# ipv6 address 2001:db8:1::1/64
```

```
Router1(config-if)# ipv6 rip MY_RIP enable
```

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

```
Router1(config)# interface Serial 0/0/0
```

```
Router1(config-if)# ipv6 address 2001:db8:100::1/64
```

```
Router1(config-if)# ipv6 rip MY_RIP enable
```

```
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
```

```
Router1(config)# do write memory
```

```
! Router 2 Configuration
```

```
Router2> enable
```

```
Router2# configure terminal
```

```
Router2(config)# ipv6 unicast-routing
```

```
Router2(config)# ipv6 router rip MY_RIP
```

```
Router2(config-rtr)# exit
```

```
Router2(config)# interface FastEthernet 0/0
Router2(config-if)# ipv6 address 2001:db8:2::1/64
Router2(config-if)# ipv6 rip MY_RIP enable
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface Serial 0/0/0
Router2(config-if)# ipv6 address 2001:db8:100::2/64
Router2(config-if)# ipv6 rip MY_RIP enable
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# do write memory
```

! Verification Command for RIPv6:

```
Router1# show ipv6 route
Router1# ping 2001:db8:2::10
```

! Task 2: Configure EIGRP for IPv6 (Same AS Number on 2 Routers)

! Topology: 2 Routers in line, each with 2 switches (Fa0/0, Fa0/1), 2 PCs/switch.

! Router 1 Configuration

```
Router1> enable
Router1# configure terminal
Router1(config)# ipv6 unicast-routing
Router1(config)# ipv6 router eigrp 10
Router1(config-rtr)# router-id 1.1.1.1
Router1(config-rtr)# no shutdown
Router1(config-rtr)# exit
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ipv6 address 2001:db8:10::1/64
Router1(config-if)# ipv6 eigrp 10
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface FastEthernet 0/1
Router1(config-if)# ipv6 address 2001:db8:11::1/64
Router1(config-if)# ipv6 eigrp 10
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Serial 0/0/0
Router1(config-if)# ipv6 address 2001:db8:100::1/64
Router1(config-if)# ipv6 eigrp 10
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# do write memory
```

```
! Router 2 Configuration
Router2> enable
Router2# configure terminal
Router2(config)# ipv6 unicast-routing
Router2(config)# ipv6 router eigrp 10
Router2(config-rtr)# router-id 2.2.2.2
Router2(config-rtr)# no shutdown
Router2(config-rtr)# exit
Router2(config)# interface FastEthernet 0/0
Router2(config-if)# ipv6 address 2001:db8:20::1/64
Router2(config-if)# ipv6 eigrp 10
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface FastEthernet 0/1
Router2(config-if)# ipv6 address 2001:db8:21::1/64
Router2(config-if)# ipv6 eigrp 10
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface Serial 0/0/0
Router2(config-if)# ipv6 address 2001:db8:100::2/64
Router2(config-if)# ipv6 eigrp 10
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# do write memory
```

! Communication Check:

! From PC on Switch 1 (IP 2001:db8:10::10), ping PC on Switch 3 (IP 2001:db8:20::10)

! Task 3: Configure OSPFv3 for IPv6 (Serial = Backbone Area 0, FastEthernet = Non-Backbone)

! Topology: 2 Routers in line, each with 2 switches (Fa0/0, Fa0/1).

! Router 1 Configuration

```
Router1> enable
Router1# configure terminal
Router1(config)# ipv6 unicast-routing
Router1(config)# ipv6 router ospf 1
Router1(config-rtr)# router-id 1.1.1.1
Router1(config-rtr)# exit
Router1(config)# interface FastEthernet 0/0
Router1(config-if)# ipv6 address 2001:db8:10::1/64
Router1(config-if)# ipv6 ospf 1 area 1
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
Router1(config)# interface FastEthernet 0/1
Router1(config-if)# ipv6 address 2001:db8:11::1/64
Router1(config-if)# ipv6 ospf 1 area 1
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Serial 0/0/0
Router1(config-if)# ipv6 address 2001:db8:100::1/64
Router1(config-if)# ipv6 ospf 1 area 0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# do write memory
```

! Router 2 Configuration

```
Router2> enable
Router2# configure terminal
Router2(config)# ipv6 unicast-routing
Router2(config)# ipv6 router ospf 1
Router2(config-rtr)# router-id 2.2.2.2
Router2(config-rtr)# exit
Router2(config)# interface FastEthernet 0/0
Router2(config-if)# ipv6 address 2001:db8:20::1/64
Router2(config-if)# ipv6 ospf 1 area 2
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface FastEthernet 0/1
Router2(config-if)# ipv6 address 2001:db8:21::1/64
Router2(config-if)# ipv6 ospf 1 area 2
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface Serial 0/0/0
Router2(config-if)# ipv6 address 2001:db8:100::2/64
Router2(config-if)# ipv6 ospf 1 area 0
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# do write memory
```

! Communication Check:

! Run: do show ipv6 ospf neighbor

! From PC on Switch 1, ping PC on Switch 4 (IP 2001:db8:21::10)

12. SWITCHING: VLAN CONFIGURATION

```
-----
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name hpcai10
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name hpcai20
Switch(config-vlan)# exit

Switch(config)# interface range FastEthernet 0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range FastEthernet 0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit

Switch(config)# do show vlan brief
Switch(config)# do write memory
```

13. SWITCHING: VLAN TRUNKING & STP

```
Switch> enable
Switch# configure terminal
Switch(config)# interface FastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
Switch(config)# spanning-tree vlan 1 priority 4096
Switch(config)# exit
Switch# do write memory
```

```
=====
=====
```